Lab03

Socket programming practice with sample programs

BUPT/QMUL

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Electronic Engineering

Tasks for Lab03

- **Task1:** Practice all the sample programs in Week2 and Week4 lectures
  - fork1.c
  - fork2.c
  - exec1.c
  - lseek1.c
  - readwrite1.c
  - udpechoclt.c / udpechosvr.c

Lab04
Network Programming with UDP

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要可以输入多个数据

server收到以后打印一些信息，比如数据来源和数据类型

Lab on UDP

- Write the programs for data sending based on UDP
  - You can design the running command format, e.g.,
    - ./<exefile> <server> <data1> <data2> ……
    - The server is specified in an IP address
    - ***Multiple words*** can be delivered each time
  - Each time the server receives data, ***the source (client's IP address and port number) and the data should be displayed on the screen***

Note: You can complete this lab by modifying the code of *UdpEchoClt.c* and *UdpEchoSvr.c* ***to support multiple words transmission***. (Example is shown in the following)

An Example for your work

- Only an example, you can design the format by yourself

1 BUPTIA | 2 BUPTIA

```
student@BUPTIA:~/example$ ./UDPsvr
./UDPsvr: waiting for data on port UDP 1500
./UDPsvr: from 127.0.0.1:UDP35666 : hello,
./UDPsvr: from 127.0.0.1:UDP35666 : who
./UDPsvr: from 127.0.0.1:UDP35666 : are
./UDPsvr: from 127.0.0.1:UDP35666 : you?
```

1. Run the server program in Terminal 1

3. Server receives the content and print it out.

1 BUPTIA | 2 BUPTIA

```
student@BUPTIA:~/example$ ./UDPclt 127.0.0.1 hello, who are you?
./UDPclt: sending data to '127.0.0.1' (IP : 127.0.0.1)
student@BUPTIA:~/example$
```

Input “IP” + “Content”

2. Run the client program in Terminal 2

TIPs

- DO NOT forget the ***<u>#include <xxx.h></u>***
- Comments in ***/* xxx */*** can be ignored
- It is SEMICOLON "；" but NOT COLON "：" in the end of each line
- Open two Xshell terminals or use ***Alt*** + F1/F2 to switch terminals in Virtualbox.
- Run the ***Server*** firstly. Then run the ***Client*** and send data
- You can use "ifconfig" to check your IP address or use "***127.0.0.1***" as your IP address when you test the program
- ***Try to read the compiling information when debug. Do not always beg help without any reading, thinking and analyzing.***

Lab04验收

- 验收范围：全体，个人验收
- 验收截止时间：下次实验课（预计第9周）

选做Lab
Finding DNS Information

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Tasks

- Write a program to find the DNS information about a given host
  - The host may be specified in domain name or IP address, e.g.,
    - ./<exefile> www.baidu.com
    - ./<exefile> 8.8.8.8
  - Use *gethostbyaddr()* and *gethostbyname()*
  - Your program shall list the official name, all the aliases, all the IP addresses in numbers-and-dots format
  - You can check your results by comparing with the output of "*nslookup*" command

一些提示

- 如何判断是根据domain name查IP还是根据IP查domain name？
  - Inet_aton()的返回值
- DNS查询后返回的是指向一个struct hostent的指针，如何引用struct中的分量？
  - Ptr->h_length
- 如何访问到alias列表和IP地址列表中的每个值？
  - for (p=ptr->h_alias; *p!=NULL; p++)
- 如何打印IP地址？
  - Inet_ntoa ()或inet_ntop ()
- 段错误

How to print IP address?

- Conversion between binary data (unsigned long) and dot-decimal notation

```
#include <arpa/inet.h>

char * inet_ntoa( struct in_addr in)

int inet_aton( const char * cp , struct in_addr * inp )
```

Only support IPv4

string(dot-decimal)

network address(binary)

1- successful
0-failed